Akoizo Original Practice - Form 3

Division B | Elastic Launched Gliders | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) If pull distance doubles, about how does stored energy change?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

A. It roughly quadruples

B. It doubles

C. It halves

D. No change

Answer: _____

2. (2 points) Which glider mass is legal?

Unit: Rules, scoring, and planning five flights.
Worked-case data: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?
Separate investigation observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

A. 6.2 g

B. 3.5 g

C. 10.0 g

D. 12.0 g

Answer: _____

3. (2 points) Two gliders collide. What may the team do?

Unit: Launch technique and the ceiling.
Worked-case data: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.
Separate investigation observations:
Practice room: 12 m ceiling; best pull 40 cm.
Competition room: 6 m ceiling with beams at 5.5 m.
First trim flight: 40 cm pull; glider hit a beam.

A. Add 5 seconds

B. Score the longer flight twice

C. Elect a re-flight after the glider lands

D. Nothing

Answer: _____

4. (2 points) Why should the launch roll and glide circle go the same direction?

A. It is required

B. It reduces mass

C. The transition is smoother without reversing the turn

D. It increases launch energy

Answer: _____

5. (4 points) Explain why sink rate, not glide ratio alone, predicts time aloft.

Answer: _____
_____
_____

6. (4 points) How can peak height be estimated consistently?

Unit: The flight log and competition day.
Worked-case data: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?
Separate investigation observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

Answer: _____
_____
_____

7. (4 points) Why design the nose early?

Unit: Construction: light, stiff, and legal.
Worked-case data: A glider weighs 3.4 g and its CG is behind the target. What should the team do?
Separate investigation observations:
Flights 1-4: 22-24 s each.
Flight 5: Hard landing on nose.
Flights 6-8: 9-12 s; glider veers right at launch.

Answer: _____
_____
_____

8. (4 points) Why must wings be stiff at launch?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

Answer: _____
_____
_____

9. (4 points) How is a tie broken?

Unit: Rules, scoring, and planning five flights.
Worked-case data: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?
Separate investigation observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A glider climbs well but loops over at the top and then dives. Suggest a fix.

Answer: _____
_____
_____

12. (2 points) Which glider has the longer glide?

Supplied synthetic observations:
Glider A: Peak 10 m; sink 0.55 m/s.
Glider B: Peak 8 m; sink 0.35 m/s.
Question: Which glides longer, ignoring climb time?

A. Glider A

B. Equal

C. Glider B

Answer: _____

13. (2 points) What does the log show about Flight 4?

Supplied synthetic observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

A. The elastic was worn out

B. It had a good launch but a poor glide trim

C. It had a poor launch

Answer: _____

14. (2 points) Define “Sheet balsa wing” in the context of Elastic Launched Gliders.

Answer: _____
_____
_____

15. (2 points) Define “Stiffness (k)” in the context of Elastic Launched Gliders.

Answer: _____
_____
_____

16. (2 points) Define “Final score” in the context of Elastic Launched Gliders.

Answer: _____
_____
_____

17. (2 points) Define “Re-flight” in the context of Elastic Launched Gliders.

Answer: _____
_____
_____

18. (2 points) Define “Banked launch” in the context of Elastic Launched Gliders.

Answer: _____
_____
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